

Maximizing CO₂ biofixation of *Arthrospira platensis* using an in situ bicarbonate pool: Optimization of CO₂-regulated pH and bicarbonate concentration

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ABSTRACT

The highly alkaline environment required for *Arthrospira* cultivation has great potential for CO₂ capture and utilization, but less attention has been paid. Here, we used *Arthrospira* medium as an in situ carbon pool to improve biomass production, carbon fixation, and capture efficiencies. Under optimized pH 8.5–10.5 and 8.4 g L⁻¹ sodium bicarbonate, the highest carbon fixation rates of *Arthrospira platensis* reached 295.68–304.75 mg L⁻¹ d⁻¹ and 26.35 g m⁻² d⁻¹ in a 3-L reactor and 5 m² runway pond, respectively, and produced biomass with abundant of phycocyanin and carotenoids. Furthermore, carbon fixation rates of 24.73–26.93 g m⁻² d⁻¹ and exogenous CO₂ utilization efficiencies of 72.62–78.24 % were obtained by culture in a large-scale 200-m² runway pond. Overall, this study established an in situ carbon pool system suitable for *Arthrospira* for carbon capture and fixation, which effectively promotes the use of *Arthrospira* in carbon emission reduction.

1. Introduction

The annual global emissions of carbon dioxide (CO₂), predominantly derived from the combustion of fossil fuels, such as petroleum, coal, and natural gas, are projected to be approximately 35 Gt [1]. This results in serious environmental issues, including the greenhouse effect and ocean acidification [2,3]. In recent years, microalgae CO₂ biofixation has emerged as a promising strategy for effective CO₂ mitigation to alleviate this issue [4–10].

Microalgae, including cyanobacteria, have received considerable attention in the fields of carbon sequestration and emissions reduction because of their efficient photosynthetic capabilities and excellent inorganic carbon conversion efficiencies [11,12]. Generally, these photoautotrophic groups directly utilize CO₂ and bicarbonate (HCO₃⁻) for photosynthesis [9,13]. Supplying gaseous CO₂ from flue gas not only regulates pH, but also enhances the availability of inorganic carbon in the medium. However, the concentration of dissolved carbon derived from CO₂ gas dissolution is limited by its low solubility [14–16]. Consequently, the low absorption ratio during the process of CO₂

supplementation is unable to provide sufficient CO₂ for microalgal growth and is also unfavorable for carbon capture from flue gas [14,15].

The cyanobacterium *Arthrospira platensis*, which contains high-quality proteins, is considered a food of the future [17,18]. *A. platensis* is suitable for growth in highly alkaline environments with high carbon concentrations. Sodium bicarbonate (NaHCO₃) is an effective carbon source for the growth of *A. platensis* due to its high solubility [9,19]. A single application of NaHCO₃ elevates the pH due to HCO₃⁻ utilization by microalgae, resulting in the release of hydroxide and limiting microalgal biomass productivity [4,15,20]. Several studies have shown that absorption rates of CO₂ increase with increasing pH and high concentrations of HCO₃⁻, which serves as a carbon pool [21–23]. The HCO₃⁻ pool plays an important role in efficient CO₂ absorption [24,25]. When absorbed by an alkaline solution, gaseous CO₂ is transformed into HCO₃⁻, which is more readily retained within the liquid medium for microalgal utilization. Based on the above findings, CO₂ biofixation of *Arthrospira platensis* by an in situ HCO₃⁻ pool is a perfect strategy to enhance exogenous CO₂ capture efficiency and reduce the use of NaHCO₃.

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The pH and HCO₃⁻ pool concentration in the medium are two important factors that significantly affect the growth and carbon fixation efficiency of *A. platensis*, and are the main determinants of exogenous CO₂ capture efficiency [26,27]. The pH of the culture solution is an important environmental factor that influences protein levels in microalgae [15] and the distribution of dissolved inorganic carbon (DIC) species in the culture medium [15]. Our previous studies confirmed that the pH and carbon concentration in the medium are important factors affecting the CO₂ absorption rate [23]. However, little attention has been paid to the combined effects of CO₂-regulated pH and inorganic carbon (NaHCO₃) concentration on the CO₂ fixation rate and exogenous CO₂ capture efficiency of *A. platensis*, as well as the production of high-value products.

In this study, we propose an in situ CO₂ capture process that enhanced carbon fixation of *A. platensis* through CO₂-regulated pH in a HCO₃⁻-based carbon pool and aimed to provide reliable culture system for combining algal CO₂ emissions and biomass production. The experimental workflow and apparatus configuration are schematically illustrated in Fig. 1. The effects of different pH levels and HCO₃⁻ pool concentrations were investigated on the carbon fixation rate, biomass yield, and production of high-value products by *A. platensis* in a 3-L columnar photobioreactor. The optimized conditions were further evaluated in 40 L and 5 m² outdoor raceway ponds for *A. platensis* strains. Subsequently, the excellent performance of the in situ carbon pool proposed and optimized in this study for carbon sequestration and biomass production was further validated in a 200 m² raceway pond.

2. Materials and methods

2.1. Microalgae strains

A. platensis strains with excellent growth performance were obtained from the Algae Culture Collection of the Wuhan Botanical Garden, Chinese Academy of Science. These strains were cultured in a modified Zarrouk's medium (Table S1). Seed cells were cultivated in 1 L flasks with 500 mL medium. The culture conditions included a light intensity of 200 μmol m⁻² s⁻¹ with a 14/10 h light/dark cycle, at a constant temperature of 30 ± 1 °C. When *A. platensis* cells were in the logarithmic growth stage, they were collected as seeds for the experiments.

2.2. pH control system

In the indoor experiments, 99.9 % CO₂ (v/v) was used for pH control, whereas 15 % CO₂ (v/v) derived from a mixture consisting of 85 % pure CO₂ (v/v) and 15 % air (v/v) was used for pH control in the outdoor experiments. The pH was monitored and adjusted using a CO₂-pH feedback control system, as described by [4].

2.3. HCO₃⁻ pool concentration and pH optimization under laboratory conditions

A bubbled column photo-bioreactor made of plexiglass was used in the experiment. Each column had an inner diameter of 15 cm and a working volume of 3 L. The columns were illuminated from two sides by cool white fluorescent tubes that provided a light intensity of 400 μmol m⁻² s⁻¹ at the reactor surface. The photoperiod was a light/dark cycle of 14 h/10 h. A temperature of 30 ± 1 °C in culture broth was maintained by a thermostatic water circulator (Hanon FCL6-20, China). Each column was aerated with an air flow of 250 mL min⁻¹ through a glass tube. Modified Zarrouk's medium was prepared using deionized water. Cells were inoculated into sterilized modified Zarrouk's medium. The initial optical density at 560 nm (OD₅₆₀) of the batch culture was 0.3 ± 0.05. Each treatment was conducted in triplicate.

To investigate the effects of HCO₃⁻ pool concentration (NaHCO₃) on cell growth and CO₂ biofixation rates, different doses of NaHCO₃ were added to each column under a CO₂-regulated pH of 9.5 before inoculation. The different concentrations of NaHCO₃ in the medium are listed in Table 1.

The effects of different CO₂-regulated pH levels on cell growth and CO₂ biofixation rates under a NaHCO₃ concentration of 8.4 g L⁻¹ were also investigated. The pH values are listed in Table 1.

2.4. Cultivation in 40 L and 5 m² outdoor open reactors using optimized carbon pool conditions

Ten *A. platensis* strains were tested in 40-L outdoor open reactors for growth characteristics and CO₂ fixation capabilities under natural light illumination and temperature conditions in Chenghai, Yunnan Province, China. Modified Zarrouk's medium was prepared using water of Chenghai Lake. The water was purified by filtration and the final medium was obtained by adding sterile nutrient stock solutions. The concentration of

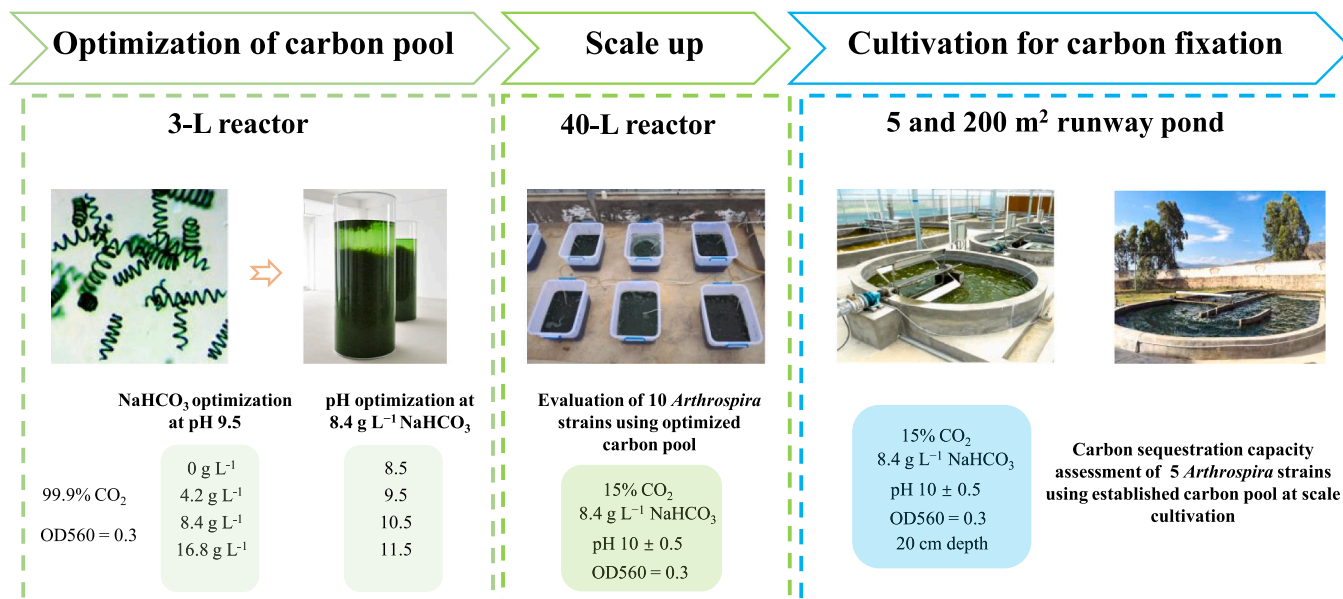


Fig. 1. Schematic diagram of experimental design, procedure, and conditions in this study.

Table 1

Biomass productivity, biochemical composition and CO₂ fixation rate of *Arthrospira* under various culture conditions. Letters noted significant differences among the treatment groups ($p < 0.05$).

NaHCO ₃ (g L ⁻¹)	pH	Biomass productivity (g L ⁻¹ d ⁻¹)	N content (%)	C content (%)	Protein content (%)	PC (mg g ⁻¹)	CO ₂ fixation rate (mg L ⁻¹ d ⁻¹)
0	9.50 ± 0.05	0.15 ± 0.01 b	10.59 ± 0.66 b	46.61 ± 2.00 b	66.17 ± 4.12 b	193.69 ± 29.54 b	255.22 ± 0.99 c
4.2	9.50 ± 0.05	0.17 ± 0.01 a	10.88 ± 0.13 a	46.92 ± 0.53 b	68.03 ± 0.79 a	219.38 ± 21.40 b	289.6 ± 4.33 b
8.4	9.50 ± 0.05	0.17 ± 0.01 a	11.26 ± 0.15 a	48.40 ± 1.42 a	70.37 ± 0.95 a	230.44 ± 3.42 a	300.98 ± 2.05 a
16.8	9.50 ± 0.05	0.17 ± 0.01 a	9.94 ± 0.23 b	48.67 ± 0.30 a	62.14 ± 1.43 c	200.41 ± 0.01 b	300.40 ± 10.15 a
8.4	8.50 ± 0.05	0.17 ± 0.01 a	11.16 ± 0.10 a	49.18 ± 1.37 a	69.77 ± 0.62 ab	214.73 ± 2.36 b	304.75 ± 4.81 a
8.4	9.50 ± 0.05	0.17 ± 0.01 a	11.26 ± 0.15 a	48.40 ± 1.42 a	70.37 ± 0.95 ab	230.44 ± 3.42 a	300.98 ± 2.05 a
8.4	10.50 ± 0.05	0.16 ± 0.01 a	11.48 ± 0.05 a	50.40 ± 0.27 a	71.78 ± 0.29 a	229.29 ± 6.28 a	295.68 ± 7.08 a
8.4	11.50 ± 0.05	0.14 ± 0.01 c	10.85 ± 0.25 a	48.48 ± 0.58 a	67.79 ± 1.55 c	218.30 ± 4.16 b	231.38 ± 10.37 c

NaHCO₃ was 8.4 g L⁻¹. The algal suspension was aerated during cultivation and the airflow was kept at 3 L min⁻¹. To maintain the culture pH at 10 ± 0.5, air enriched with CO₂ (15 %, v/v) was injected into the culture solution through an aeration header (diameter: 30 mm, pore size: 1 mm) in the 40 L reactors. The 15 % CO₂ (v/v) was obtained from a blend of pure CO₂ (≥ 99.9 %) and air. The initial OD₅₆₀ was 0.3 ± 0.05. The dry weight (DW) of the biomass was measured daily.

Five *A. platensis* strains were selected and tested for their growth characteristics and CO₂ fixation rates in 5 m² outdoor open reactors. The five open outdoor reactors used in this experiment were located in Chenghai, Yunnan Province, China. The reactor was constructed using concrete blocks. Each reactor had an illuminated area of 5 m² and was circular. A paddlewheel with six blades was used for mixing and was driven by a 0.75 kW electric motor. The liquid flow velocity was set as 45–50 cm s⁻¹ by adjusting the paddlewheel speed, which is conducive to the algal growth and the mass transfer of exogenous CO₂. The other culture conditions were similar to those used in the 40-L outdoor open reactors. The DW of the biomass was measured daily. Natural solar irradiance and environmental temperatures were also monitored.

2.5. Biomass production and carbon sequestration by *A. platensis* strain GT-1 in a 200 m² open raceway pond

A pilot-scale evaluation of biomass production and carbon sequestration by *A. platensis* strain GT-1 in an optimized carbon pool was conducted in a classical raceway pond. This reactor was located at Chenghai Lake, Yunnan province, China, (N26°29'29.64" E100°40'56.12"). This reactor (20 m long and 12 m wide) was an open system with an illuminated area of 200 m². Two large paddle wheels were used for mixing and were driven by an electric motor. The liquid flow velocity was set at a desired range of 45–50 cm s⁻¹. Air enriched with CO₂ (15 %, v/v) was automatically injected into solution using a gas diffuser. To enhance the CO₂ absorption ratio, a 3-m long micro-porous tube was placed at the bottom of the culture pond. The pH was regulated using 15 % CO₂ (v/v) to maintain the pH of the culture solution within the desired range (pH 10 ± 0.5). The working culture volume was 40,000 L (20 cm depth). Two independent batch cultures in the 200 m² pond were conducted.

2.6. Analytical procedure

Cell growth and biomass were estimated by determining biomass DW of the algal suspension. The DW was determined by filtering 5 mL of algal suspension through a pre-weighed nitrocellulose filter membrane with a pore size of 0.45 μm, followed by drying at 105 °C for 4 h. The areal biomass production (B, g m⁻²) of the outdoor culture was calculated using the following equation:

$$B = \frac{DW \times V}{S} \quad (1)$$

Where V represents the culture volume in liters (L), and S represents the illuminated area in square meters (m²).

Biomass productivity (P_B, g m⁻² d⁻¹) was calculated using the following equation:

$$P_B = \frac{B_{t1} - B_{t0}}{t1 - t0} \quad (2)$$

Where B_{t1} and B_{t0} represent biomass production at culture time t₁ (day) and t₀ (day), respectively.

Carotenoid levels were determined using the method described by [28]. The carotenoid content (C_{carotenoid}, mg/L) was calculated using the following equation:

$$C_{carotenoid} = (1000 \times OD_{470} - 1.63 \times Chla) / 221 \quad (3)$$

Where Chla represents the chlorophyll a content.

Phycocyanin (PC) was extracted using phosphate buffer (pH 6.8) and absorbance was measured at 620 nm and 652 nm, respectively. The operational procedures have been described by [29]. The concentrations of PC in biomass were calculated according to the equation described by [30].

The photosynthetic status of *A. platensis* was measured using a pulse amplitude modulation fluorometer (PAM 2500; Heinz Walz, Effeltrich, Germany). The operational procedures were conducted as described by [20]. Fv/Fm, which represents the maximum photochemical efficiency of photosystem II (PSII) was determined at a temperature of 30 °C and calculated using the following equation:

$$Fv / Fm = \frac{F_m - F_0}{F_m} \quad (4)$$

Where Fv represents the variable fluorescence, F₀ is minimal fluorescence level, and F_m is maximal fluorescence level after the addition of 3-(3,4-dichlorophenyl)-1, L-dimethyl urea to the algal samples. Next, the sample was dark-adapted for 10 min. Algal cultures (3 mL) were sampled and placed in a light-protected measuring chamber with a light path of 10 mm. Before measurements, the culture was diluted to an OD₅₆₀ of 0.5 to avoid re-absorption problems in the dense suspension. Rapid light-response curves of relative electron transport were fitted using the formula proposed by [31].

The total carbon (TC) and nitrogen (TN) content in the algal biomass were measured using a Soli TOC cube analyzer (Elementar, Germany). The protein content was calculated using the following equation described by [32]:

$$\text{Protein content}(\%) = 6.25 \times \text{TN}(\%) \quad (5)$$

The CO₂ fixation rate (Fc) of microalgae was calculated using the equation:

$$Fc = P_B \times TC \times M_{CO_2} / M_C \quad (6)$$

Where M_{CO2} and M_C represent the molecular weights of CO₂ and C, respectively [33].

Utilization efficiency of exogenous CO₂ (R_c, %) was calculated according to the equation described by [4]:

$$Rc(\%) = \frac{B \times TC \div 12}{C_{t0} - C_t + C} \times 100\% \quad (7)$$

Where C_{t0} represents total inorganic carbon (mol) in the medium at the beginning of culture, C_t represents total inorganic carbon (mol) in the medium at the end of culture, and C represents the total CO₂ (mol) used during culture.

2.7. Statistics

Results are presented as the mean $\pm$ standard deviation. Significant differences between treatment groups were determined using one-way analysis of variance with SPSS software (version 13.0; SPSS Inc., Chicago, IL, USA), where differences were considered significant at $p < 0.05$.

3. Results and discussion

3.1. Optimization of HCO₃⁻ pool concentration (NaHCO₃) for the growth and CO₂ fixation of *A. platensis* strain HN5

The addition of NaHCO₃ to the medium had a significant effect on biomass production of *A. platensis* under a CO₂-regulated pH of 9.5 in 3-L columnar photobioreactors (Fig. 2A). On day 4, the biomass production in the cultures containing 4.2 g L⁻¹, 8.4 g L⁻¹, and 16.8 g L⁻¹ of NaHCO₃ was higher than that in cultures without NaHCO₃ ($p < 0.05$). This trend continued to the tenth day. The highest biomass production of 1.85 g L⁻¹ was observed at a concentration of 8.4 g L⁻¹ NaHCO₃ on day 10, which was 12.2 % higher than that in cultures without NaHCO₃ ($p < 0.05$). By contrast, no significant difference was observed among the treatments containing 4.2 g L⁻¹, 8.4 g L⁻¹, and 16.8 g L⁻¹ of NaHCO₃ at the different cultivation time points ($p > 0.05$). Thus, a

concentration of 8.4 g L⁻¹ NaHCO₃ was used for later experiments.

CO₂-regulated pH is an effective strategy to achieve biomass production and CO₂ fixation by microalgae [4,7,15]. Generally, the CO₂ injected into liquid medium transforms rapidly into DIC (CO₂ (aq) + HCO₃⁻ + CO₃²⁻) through a rapid chemical reaction and the available DIC (CO₂ (aq) + HCO₃⁻) is assimilated by photosynthesis in microalgae [12,23]. However, the production of available DIC is limited in the liquid medium because of the low solubility of CO₂ (g) at a CO₂-regulated pH of 9.5 and the available DIC concentrations do not meet the requirements for the higher growth rate of microalgae [14]. The addition of NaHCO₃ to the medium increases the availability of inorganic carbon (DIC, CO₂ + HCO₃⁻) for *A. platensis* growth under a CO₂-regulated pH of 9.5, due to its high water solubility [12,14,15]. Moreover, the results of our previous studies have shown that an increase in the carbon concentration within Zarrouk's medium enhances the CO₂ absorption rate, leading to improved carbon availability [23]. Thus, these culture conditions were beneficial for the growth and photosynthesis of *A. platensis*.

Photosynthesis in *A. platensis* was analyzed under different culture conditions (Fig. 2B, C, D, and E). The maximum quantum efficiency of PSII (Fv/Fm) decreased along a gradient of NaHCO₃ concentrations and exhibited a dose-dependent manner under the NaHCO₃ concentrations between 0 and 16.8 g L⁻¹ after the 10-day culture. This suggested that the addition of NaHCO₃ had an inhibitory effect on the activity of PSII in *A. platensis*. NaHCO₃ dissolved in water releases large amounts of sodium ions, which increase the salinity of the medium and significantly influence photosynthesis in *A. platensis* [34]. Thus, the quantity of NaHCO₃ that was added had to be optimized. The maximum light use efficiency (α) was significantly lower in the culture without NaHCO₃ than those with NaHCO₃ ($p < 0.05$). This may be because NaHCO₃ provides an additional source of DIC, which enhances the photosynthetic efficiency of *A. platensis* and promotes its growth [14,35]. In

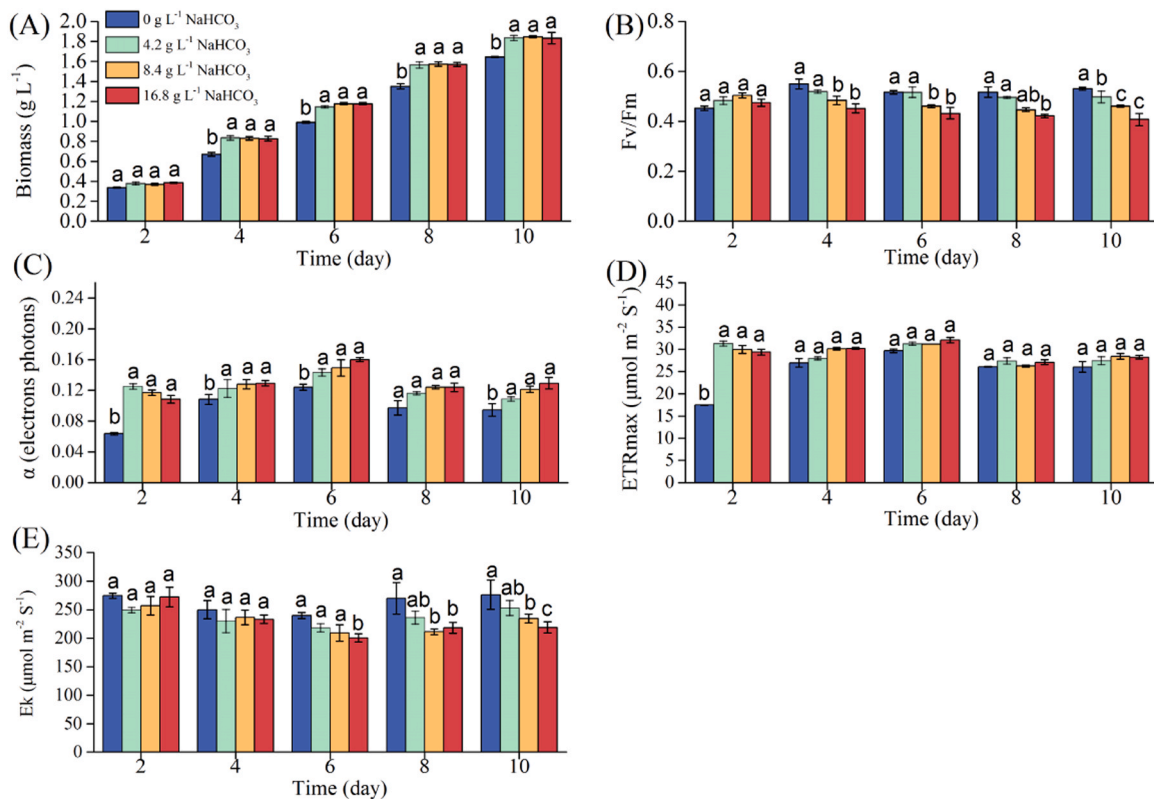


Fig. 2. Effects of sodium bicarbonate concentration on the growth and photosynthesis of *Arthrospira* sp. at pH 9.5. (A) Biomass dry weight, (B) The maximal quantum yield (Fv/Fm), (C) the maximum light use efficiency (α), (D) The maximal electron transport rate (ETRmax) (E) the light saturation coefficient (Ek). All data are expressed as mean values $\pm$ standard deviation ($n = 3$). Letters noted significant differences among the treatment groups ($p < 0.05$).

contrast, no significant differences in the maximum electron transport rate (ETR_{max}) were observed among the four treatments ($p > 0.05$). At the end of the experiment, the light saturation coefficient (E_k) significantly decreased along a gradient of NaHCO₃. These data suggest that the addition of NaHCO₃ significantly decreased the tolerance of *A. platensis* to high-intensity light. These results indicate that the addition of NaHCO₃ to the medium increased the light-use efficiency of *A. platensis*, leading to higher biomass production.

As shown in Table 1, NaHCO₃ addition to the medium enhances the CO₂ fixation rate of *A. platensis* at a CO₂-regulated pH of 9.5. This finding is consistent with those of previous studies showing that a HCO₃⁻-based cultivation system enhances the CO₂ fixation rate of microalgae under optimal CO₂-regulated pH conditions [12,14,35]. For example, Qi et al. (2019) have shown that the addition of HCO₃⁻ to the culture medium significantly increases CO₂ fixation rates, suggesting that HCO₃⁻ may be a key factor in optimizing microalgal cultivation systems for CO₂ sequestration [35]. Generally, the CO₂ fixation rate was determined and calculated using the C content and the corresponding biomass productivity of *A. platensis*. The fastest growth was observed for the 8.4 g L⁻¹ NaHCO₃ treatment, which had a final average biomass productivity of 0.17 g L⁻¹ d⁻¹. A maximum C content of 48.67 % was observed at a concentration of 16.8 g L⁻¹ NaHCO₃, which was close to that observed at a concentration of 8.4 g L⁻¹ NaHCO₃. However, both were significantly higher than those observed in the treatments with 0 or 4.2 g L⁻¹ NaHCO₃. The CO₂ fixation rate was similar to the changes in C content. A maximum CO₂ fixation rate of 300.98 mg L⁻¹ d⁻¹ was observed at a concentration of 8.4 g L⁻¹ NaHCO₃. Maximum protein content was observed for the 8.4 g L⁻¹ NaHCO₃ treatment, which had a final average protein content of 70.37 %. Similar to the protein content, the maximum PC content (230.44 mg g⁻¹) was observed at a concentration of 8.4 g L⁻¹ NaHCO₃. Therefore, it was essential to optimize the pH in a HCO₃⁻-based microalgal cultivation system with a concentration of 8.4 g L⁻¹ NaHCO₃.

3.2. pH conditions of the carbon pool that were suitable for the growth and CO₂ fixation of *A. platensis* strain HN5

As shown in Fig. 3, biomass production gradually decreases along a gradient of pH levels in the 3-L columnar photobioreactors at a HCO₃⁻ pool concentration of 8.4 g L⁻¹ NaHCO₃, indicating that elevated pH levels inhibit the growth of *A. platensis*. Cell morphology analysis supported this finding (Fig. S1). When the solution pH was 9.5, *Arthrospira* filaments were normal, whereas the filaments were severely broken when exposed to both pH 6.5 and pH 12. The maximum biomass production of 1.84 g L⁻¹ was observed at pH 8.5. No significant difference was observed among the pH 8.5, 9.5, and 10.5 treatments ($p > 0.05$). These findings are consistent with the results of our preliminary experiment (Fig. S2) and other reports [12,15,20]. This result is attributed to the significant effects of solution pH on the morphology, photosynthesis, and biochemical composition of *A. platensis* [20]. In addition, the pH of the solution determines the ratio of DICs (CO₂ (aq) + HCO₃⁻ + CO₃²⁻). According to the equilibrium formula, at pH 8.5, the main form of DIC is HCO₃⁻, which can be utilized by *Arthrospira*. However, as the pH increases, the proportion of bicarbonate rapidly decreases, and the proportion of CO₃²⁻ that cannot be utilized by *Arthrospira* rapidly increases. Therefore, the highest biomass yield obtained at pH 8.5 is maybe closely related to the highest concentration of available DIC.

The Fv/Fm gradually decreased along the pH gradient (Fig. 3B), indicating that elevated pH levels had an inhibitory effect on PSII activity in *A. platensis*. This finding is consistent with those of other studies [20]. When the Fv/Fm of *A. platensis* decreases to a certain value under pH stress, the corresponding biomass yield significantly decreases (Fig. 3, [20]). In the early stage of the experiment, the α of *A. platensis* responded rapidly to pH changes. At the end of the experiment, α in pH 9.5 treatments were significantly lower than that in other treatments. These data suggest that the light requirement of algae decreased when the CO₂-regulated pH was 9.5. This finding is likely due to the fact that a

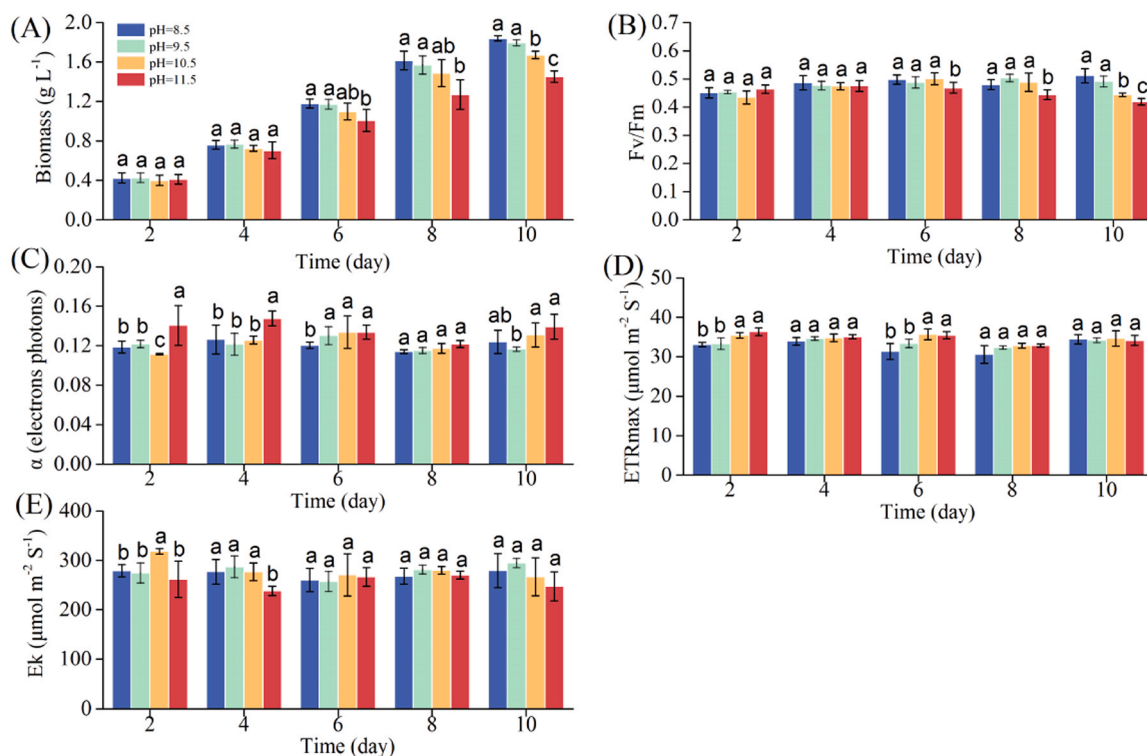


Fig. 3. Effects of the different pH levels on the growth and photosynthesis of *Arthrospira* sp. at a concentration of 8.4 g L⁻¹ NaHCO₃. (A) Biomass dry weight, (B) The maximal quantum yield (Fv/Fm), (C) the maximum light use efficiency (α), (D) The maximal electron transport rate (ETR_{max}) (E) the light saturation coefficient (E_k). All data are expressed as mean values $\pm$ standard deviation ($n = 3$). Letters noted significant differences among the treatment groups ($p < 0.05$).

solution with a pH of 9.5 provides an increased carbon source, thereby reducing the reliance on light energy [20]. In contrast, no significant effects on the ETR_{max} or E_k were observed among the four pH levels tested ($p > 0.05$). These findings indicate that the pH of the solution did not influence the ETR_{max} or the high light intensity tolerance of *A. platensis* within a pH range of 8.5–11.5. Therefore, the pH of the solution was maintained within a range of 8.5–10.5, which was the optimal culture condition for *A. platensis* photosynthesis.

As shown in Table 1, the biomass productivity in the pH 8.5 treatment is $0.17 \text{ g L}^{-1} \text{ d}^{-1}$, which is the same as those in the pH 9.5 and 10.5 treatments. However, these values were higher than those observed at pH 11.5. The pH had no effect on the C and N content of *A. platensis*. A maximum CO₂ fixation rate of $304.75 \text{ mg L}^{-1} \text{ d}^{-1}$ was observed at pH 8.5. No significant difference was observed among the pH 8.5, 9.5, and 10.5 treatments ($p > 0.05$). The observed trend in the CO₂ fixation rate was consistent with the biomass productivity. This finding indicates a positive correlation between the CO₂ fixation rate and biomass productivity in the CO₂ controlled culture ecosystem [12]. The protein content gradually increased as the pH ranged from 8.5–10.5. When the pH was > 10.5 , the protein content was significantly reduced ($p < 0.05$). No significant differences in protein content were observed among the pH 8.5, 9.5, and 10.5 treatments ($p > 0.05$). The observed trend in protein content was consistent with that of the PC content. Therefore, in this study, the optimum pH range was the range of 8.5–10.5.

3.3. Biomass production and carbon fixation of *A. platensis* strains in 40 L and 5 m² reactors at suitable carbon pool conditions

Ten *A. platensis* strains with potential for high CO₂ fixation capability were cultivated in 40-L open reactors with a concentration of 8.4 g L^{-1} NaHCO₃ under the optimum pH ranges of 9.5–10.5. Growth curves of the 10 *A. platensis* strains are shown in Fig. 4A. The 10 strains grew well in culture under optimal conditions. However, significant differences were observed in the growth of the 10 strains. At the end of cultivation, the biomass production of the 10 strains ranged from 0.99 g L^{-1} to 1.17 g L^{-1} . The highest biomass production of 1.17 g L^{-1} was for strain GT-6. The biomass production of the 10 strains was arranged in descending order as follows: GT-6, GT-1, GT-3, GT-7, GT-5, GT-15, GT-2, GT-13, GT-11, and GT-14.

The carotenoid and PC contents of the 10 strains were also measured (Fig. 4B and C). The carotenoid content of the 10 strains increased linearly over time. The highest carotenoid content for strain GT-5 was 3.5 mg L^{-1} . The PC content of the 10 strains ranged from 11.86–16.96 %. The highest PC content for strain GT-1 was 16.96 %. The 10 strains showed a significant difference in CO₂ fixation rate (Fig. 4D). The CO₂ fixation rate of the 10 strains ranged from 181.81 – $215.23 \text{ mg L}^{-1} \text{ d}^{-1}$. The CO₂ fixation rates of the 10 strains decreased in the following order: GT-6, GT-1, GT-3, GT-7, GT-15, GT-5, GT-2, GT-13, GT-14, and GT-11.

To improve carbon fixation efficiency, excellent *A. platensis* strains must be selected [12]. Generally, algal strains vary in growth rate, ability to grow in high-concentrations of NaHCO₃, morphological characteristics, ability to assimilate different forms of carbon sources, light requirements, tolerance to high CO₂ levels, pH, and cellular

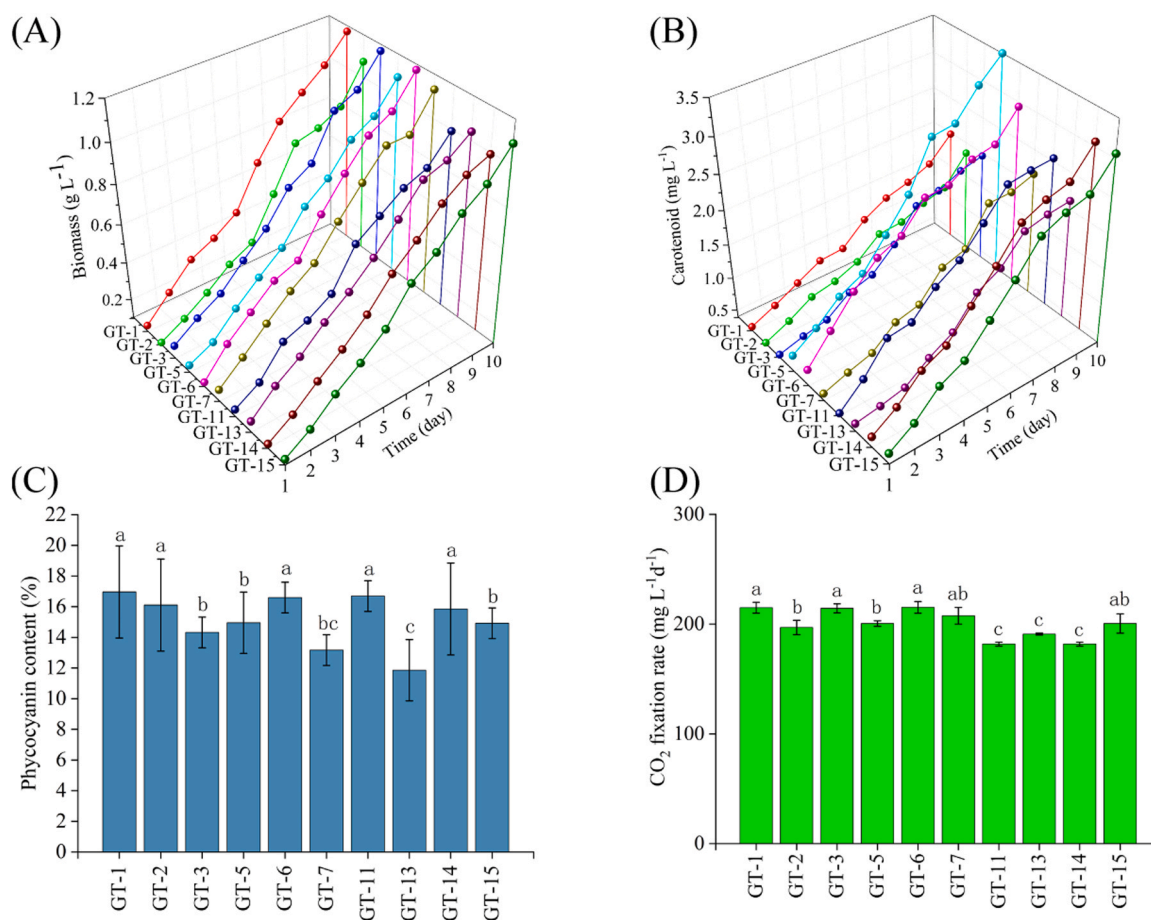


Fig. 4. Growth and carbon fixation of ten *A. platensis* strains in 40 L outdoor open reactors (A) Growth curves, (B) The carotenoid content (C) The phycocyanin content and (D) CO₂ fixation rate. All data are expressed as mean values $\pm$ standard deviation ($n = 3$). Letters noted significant differences among the treatment groups ($p < 0.05$).

composition [12,15,20,21,36–39]. For example, Zhu et al. (2021) have reported three *A. platensis* strains tested for their CO₂ fixation capabilities at different CO₂ concentrations; *A. platensis* strain LAMB220 has the best biomass productivity and C-fixation capability at a 10 % CO₂ concentration [38]. Moreover, there are significant differences in the activity of certain enzymes related to CO₂ biofixation, such as ribulose-1, 5-bisphosphate carboxylase/oxygenase, across different strains [11,40]. This may be related to the genetic characteristics and environmental adaptability of these strains [41]. These differences may affect the efficiency of photosynthesis and CO₂ fixation, which in turn affect growth and biomass accumulation of the strains [40,42]. Previous results suggest that algal strain selection may be a key strategy to improve the carbon sequestration capacity of algae [11,27]. In addition, the morphological characteristics of the 10 *A. platensis* strains were examined. Although strains GT-6, GT-7, and GT-15 have better C-fixation capabilities, they were not selected because of unsatisfactory morphological features that affect biomass harvesting efficiency [43]. Based on the above results, including the CO₂ biofixation capability and morphological characteristics of the strains, five strains were selected for culture in larger-scale outdoor open reactors.

The growth curves of five *A. platensis* strains in 5 m² open raceway reactors with a concentration of 8.4 g L⁻¹ NaHCO₃ under the optimum pH ranges of 9.5–10.5 is shown in Fig. 5A. The five strains grew rapidly over time. At the end of cultivation, the biomass production of the five strains ranged from 0.52 g L⁻¹–0.72 g L⁻¹. The highest biomass production was 0.72 g L⁻¹ in strain GT-1. The biomass production of the five strains decreased in the following order: GT-1, GT-13, GT-2, GT-11, and GT-3.

The carotenoid and PC content of the five strains was also measured (Fig. 5B and C). The highest carotenoid content for strain GT-3 was 3.78 mg L⁻¹. The PC content of the five strains ranged from

12.77–15.78 %. The highest PC content for strain GT-1 was 15.78 %. The CO₂ fixation rate of the five strains ranged from 95.90–131.78 mg L⁻¹ d⁻¹ (Fig. 5D), and decreased in the following order: GT-1, GT-13, GT-2, GT-7, and GT-3.

These results suggest that *A. platensis* strain GT-1 had the best biomass production and an advantage in terms of carbon sequestration capacity. Microalgal growth in open outdoor reactors is significantly influenced by various biotic and abiotic factors [44–46]. Biological contamination is a biotic factor that poses significant challenges for microalgal cultivation in open outdoor reactors [47]. In this study, ciliate, rotifer, and other non-target microalgal contaminations in *A. platensis* cultures were observed, but there were no adverse effects on the growth of the five strains (data not shown). In contrast, abiotic factors, such as light, temperature, and nutrients may affect the growth of *A. platensis* [12]. Thus, we speculated that the differences in the growth of the five strains were due to the combined effects of adverse abiotic factors.

3.4. Carbon fixation and utilization efficiency of *A. platensis* strain GT-1 under large-scale suitable carbon pool conditions

Based on cultivation in 40 L and 5 m² reactors, the biomass production, carbon fixation rate, and utilization efficiency of *A. platensis* strain GT-1 was systematically evaluated under suitable carbon pool conditions in a 200 m² raceway pond. Two batch cultures of *A. platensis* strain GT-1 in 200 m² raceway reactors were performed (Fig. 6). The temporal evolution of biomass production and carotenoid content during the two cultivation periods is shown in Fig. 6A and B. For the first cultivation period, *A. platensis* strain GT-1 grew rapidly from the third day in the open raceway reactor and biomass production increased from 28.51 to 161.56 g m⁻². The trend in carotenoid content over time was

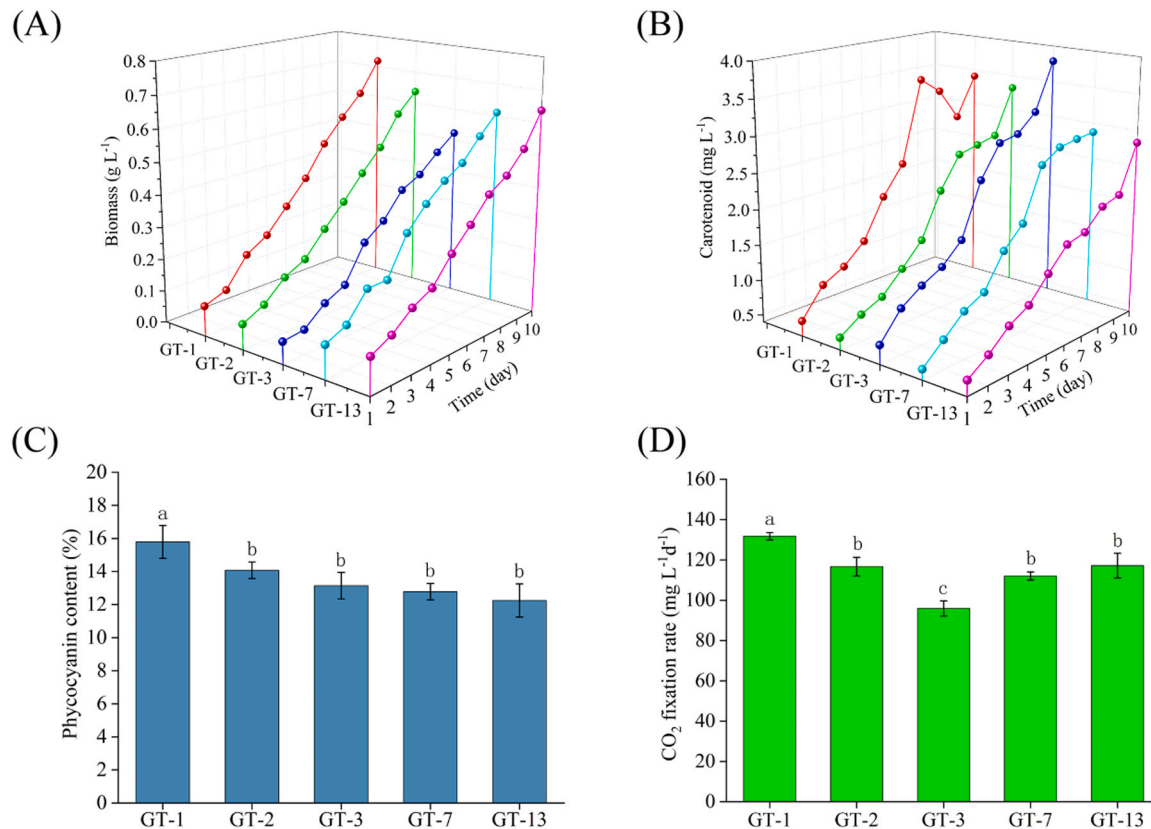


Fig. 5. Growth and carbon fixation of five *A. platensis* strains in 5 m² open raceway reactors. (A) Growth curves, (B) The carotenoid content (C) The phycocyanin content and (D) CO₂ fixation rate. All data are expressed as mean values $\pm$ standard deviation ($n = 3$). Letters noted significant differences among the treatment groups ($p < 0.05$).

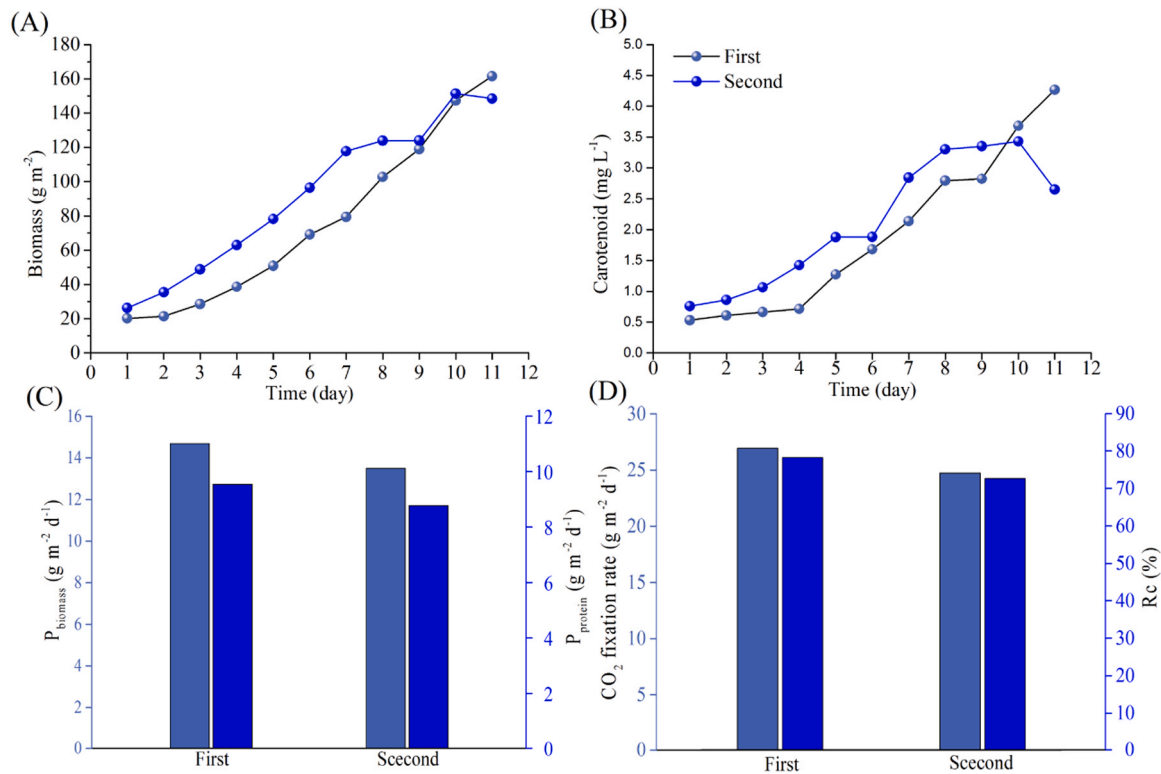


Fig. 6. Growth characteristics of *A. platensis* GT-1 in 200 m² open raceway reactors (A) Growth curve, (B) The carotenoid content, (C) Biomass productivity and protein productivity (P_{biomass} and P_{protein}) and (D) CO₂ fixation rate and utilization efficiency of exogenous CO₂ (Rc). The batch experiment was carried out twice (First and Second).

consistent with that of the biomass production and a maximum carotenoid level (4.27 mg L⁻¹) was observed at the end of cultivation. The CO₂ fixation rate, exogenous CO₂ utilization efficiency, and protein productivity were 26.93 g m⁻² d⁻¹, 72.62 %, and 9.54 g m⁻² d⁻¹, respectively. In the second cultivation, a maximum biomass production of 153.38 g m⁻² and carotenoid content of 3.43 mg L⁻¹ were observed (Fig. 6C). The CO₂ fixation rate, exogenous CO₂ utilization efficiency, and protein productivity were 24.73 g m⁻² d⁻¹, 78.24 %, and 8.77 g m⁻² d⁻¹, respectively (Fig. 6D).

The coupling of carbon fixation with valuable microalgal products is important for sustainable development [48]. These results showed that strain GT-1 exhibited robust growth and carbon fixation in a 200 m² open raceway reactor under optimal cultivation conditions (a CO₂-regulated pH range of 9.5–10.5 and 8.4 g L⁻¹ NaHCO₃), and, at the same time, showed great potential in protein and carotenoid production. The effective utilization efficiency of exogenous CO₂ was observed to be > 70 %, indicating that under these specific conditions, a HCO₃⁻-based cultivation system for *A. platensis* absorbed exogenous CO₂ with optimal efficiency. This result is even higher than that those observed in mass cultures of *Spirulina* (60 %) [49], *Phaeodactylum* (63 %) [50], *Chlorella* (46 %) [51], and at a similar level as *Graesiella* sp. WBG-1 (71.17 %) [4] (Table 2). This might be attributed to the effects of the pH and NaHCO₃ concentration. In an open raceway pond, the escape of CO₂ during aeration causes a large amount of CO₂ loss and leads to a lower utilization efficiency of exogenous CO₂ [52]. High pH prevents the escape of CO₂ from the culture. Fortunately, high pH was generally favorable for the growth of *A. platensis*. Moreover, the utilization of HCO₃⁻ by *A. platensis* regenerates carbonate, which may be re-used as an absorbent to capture CO₂ [21]. Alkaline conditions and high NaHCO₃ concentrations also facilitate the absorption of CO₂ from the air by the culture system [25]. This is because increased pH and HCO₃⁻ concentrations enhance the capacity of the *A. platensis* cultivation system to capture and utilize CO₂, thereby improving the overall CO₂ fixation rate [53]. In

Table 2

Summary of the carbon sequestration capacity of microalgae in the different reactors.

Species	Reactors	P_{biomass} (g m ⁻² d ⁻¹)	CO ₂ fixation rate (g m ⁻² d ⁻¹)	Rc (%)	References
<i>Arthrospira</i> sp. GT-1	200 m ² Raceway (First)	14.69	26.93	72.64	This study
<i>Arthrospira</i> sp. GT-1	200 m ² Raceway (Second)	13.49	24.73	78.24	This study
<i>Graesiella</i> sp. WBG-1	5 m ² Raceway	10.31	18.9	71.17	[4]
<i>Graesiella</i> sp. WBG-1	200 m ² Raceway	8.7	15.95	65.7	[44]
<i>Phaeodactylum tricornutum</i>	TAP	-	-	63	[50]
<i>Chlorella</i> sp.	A mini-raceway pond	-	-	46	[51]
<i>Spirulina platensis</i>	A mini-raceway pond	-	-	39	[51]
<i>Spirulina platensis</i>	10,000 m ² Raceway	-	-	60	[49]

Rc: Utilization efficiency of exogenous CO₂, TAP: Tubular airlift photobioreactors, -: the data was not provided in the article.

summary, the highly effective utilization efficiency of exogenous CO₂ under alkaline conditions with high NaHCO₃ concentrations highlights the importance of optimizing pH and carbon concentration to maximize the CO₂ fixation efficiency and growth of *A. platensis* in open culture systems.

4. Conclusions

The cultivation medium of *Arthrospira* showed great potential as an in situ carbon pool to promote carbon sequestration and biomass production, under optimized 8.4 g L⁻¹ NaHCO₃ and pH 8.5–10.5. In 40 L and 5 m² reactors, the highest carbon fixation rate reached 215.23 and 131.78 mg L⁻¹ d⁻¹, respectively, under suitable carbon pool conditions. At the level of large-scale cultivation (a 200 m² raceway pond), the carbon fixation rate of 24.73–26.93 g m⁻² d⁻¹ and exogenous CO₂ utilization efficiencies of 72.62–78.24 % were obtained. The in situ carbon pool technology established in the present study significantly enhanced carbon sequestration and biomass production of *A. platensis*.

CRediT authorship contribution statement

Zhongjie Wang: Resources, Funding acquisition, Supervision, Writing – review & editing, Methodology, Conceptualization. **Yi Ding:** Methodology, Resources, Data curation. **Kaixuan Wang:** Investigation, Methodology, Formal analysis, Data curation. **Yan Xu:** Methodology, Formal analysis, Investigation, Data curation. **Youzhi Yu:** Writing – original draft, Resources, Formal analysis, Conceptualization, Writing – review & editing, Validation, Investigation, Visualization, Methodology, Data curation. **Yeguang Li:** Resources, Supervision, Conceptualization, Writing – review & editing, Methodology. **Yahong Geng:** Methodology, Data curation, Formal analysis, Investigation. **Xiaobin Wen:** Investigation, Conceptualization, Methodology, Data curation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jcou.2025.103172](https://doi.org/10.1016/j.jcou.2025.103172).

Data availability

Data will be made available on request.

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